

3D Deep Learning 67801 - Exercise 2 | Problem 1 - Report

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A. Short-Answer Questions

1. Why are the MLP reconstructions so much less detailed than those produced by the SIREN?

ReLU MLPs suffer from **spectral bias**: they learn low-frequency, smooth structure in (x, y) much faster than the high frequencies needed for edges and fine texture. Because ReLU layers are piecewise linear, the fitted field stays largely piecewise linear over coordinate space and cannot efficiently represent sharp or rapidly varying detail. SIRENs use sine activations with ω_0 and SIREN-specific initialization so the network can represent a broad range of spatial frequencies from early training, better matching an image as a neural field and yielding sharper reconstructions.

2. The image Laplacian produced by the MLP looks strange... what's happening here and why?

This happens because a ReLU network is **piecewise linear** in coordinates: the Laplacian is zero wherever activations are fixed, with sparse spikes only at ReLU boundaries—so the panel highlights representation artifacts rather than semantic edges.

B. Hyperparameter sweep and model selection

Shared settings for all runs: `ImageDataset(height=128), total_steps=2000, steps_til_summary=500, set_seed(0)`, Adam optimizer, MSE loss. SIREN runs use `hidden_layers=3` throughout; default head is **linear last** (`last_layer_linear=True`, paper-style), with a few sine-last runs for comparison.

For each sweep we list final PSNR in a table, give a short analysis per configuration, show field and training curves for selected runs, and include a bar chart comparing PSNR across all runs in that sweep.

1. MLP sweep

Run	Width	Hidden layers	Activation	LR	Notes	PSNR (dB)
M1	128	3	ReLU	10^{-4}	Baseline	10.46
M2	256	3	ReLU	10^{-4}	Wider	12.02
M3	128	6	ReLU	10^{-4}	Deeper	13.00
M4	128	3	Tanh	10^{-4}	Smooth activation	8.62
M5	128	3	ReLU	10^{-3}	LR high	12.31
M6	128	3	ReLU	10^{-5}	LR low	6.72

M1 — Baseline (10.46 dB) This is our reference ReLU MLP: blurry reconstruction and a Laplacian that spikes on ReLU boundaries rather than on true image edges, which is typical **spectral bias** at this resolution and step budget. Later runs are judged relative to this PSNR and these visuals.

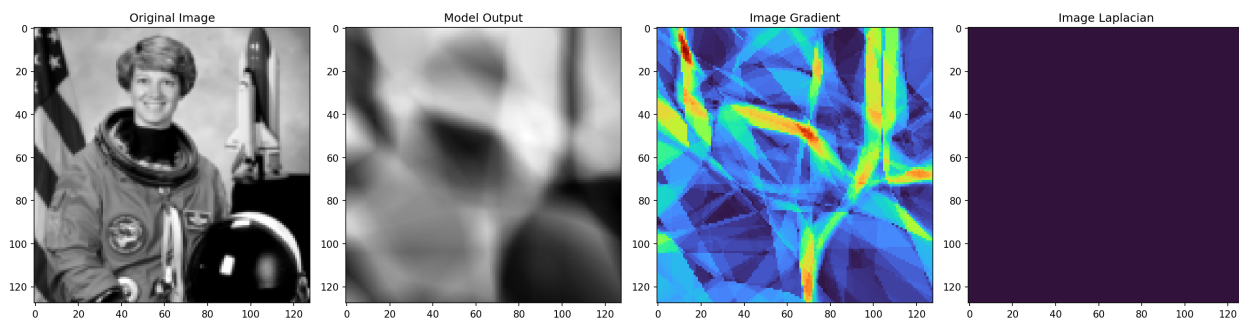


Figure 1: M1 field at step 1999

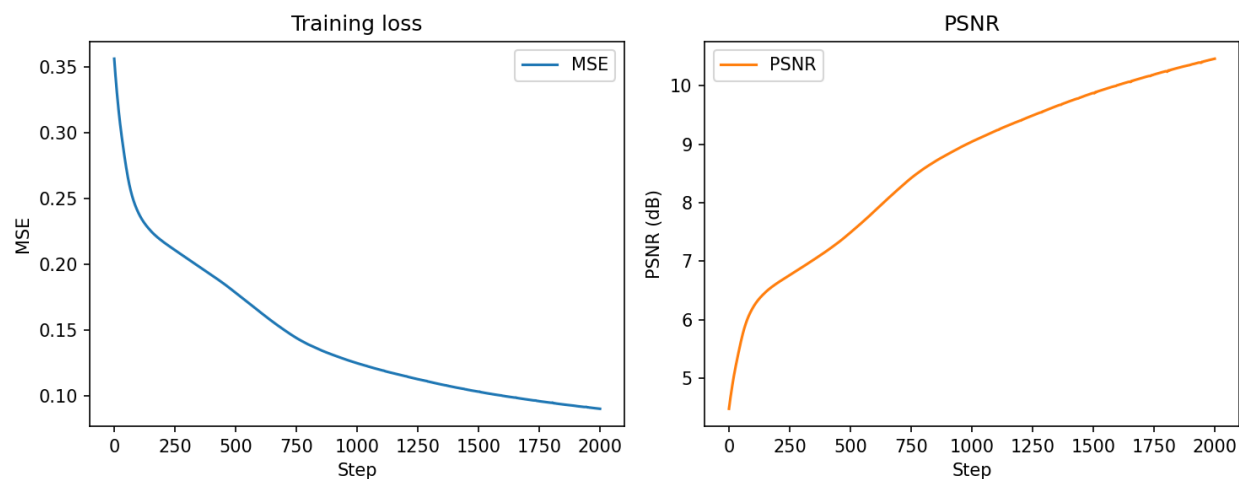


Figure 2: M1 metrics at step 1999

M2 — Wider (12.02 dB, +1.56 vs M1) Doubling hidden_features adds capacity: more ReLU “pieces” over (x, y) , so the field can approximate slightly sharper variation. PSNR improves, but the reconstruction is still smooth compared with SIREN because the **inductive bias** (piecewise-linear ReLU) is unchanged.

M3 — Deeper (13.00 dB, +2.54 vs M1) Depth helps more than width here (**best MLP**, 13.00 dB), with gradients that better follow image edges, but the reconstruction stays blurry next to SIREN. The Laplacian panel is flat, as expected for a piecewise-linear ReLU field (zero second derivative almost everywhere).

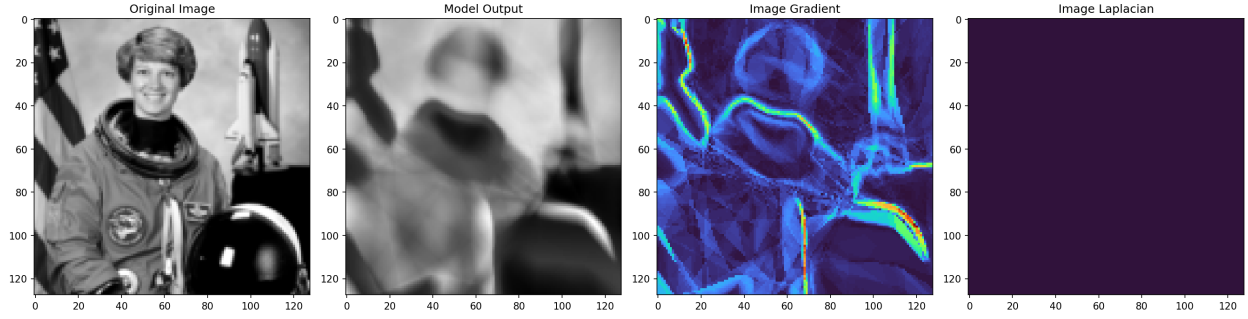


Figure 3: M3 field at step 1999

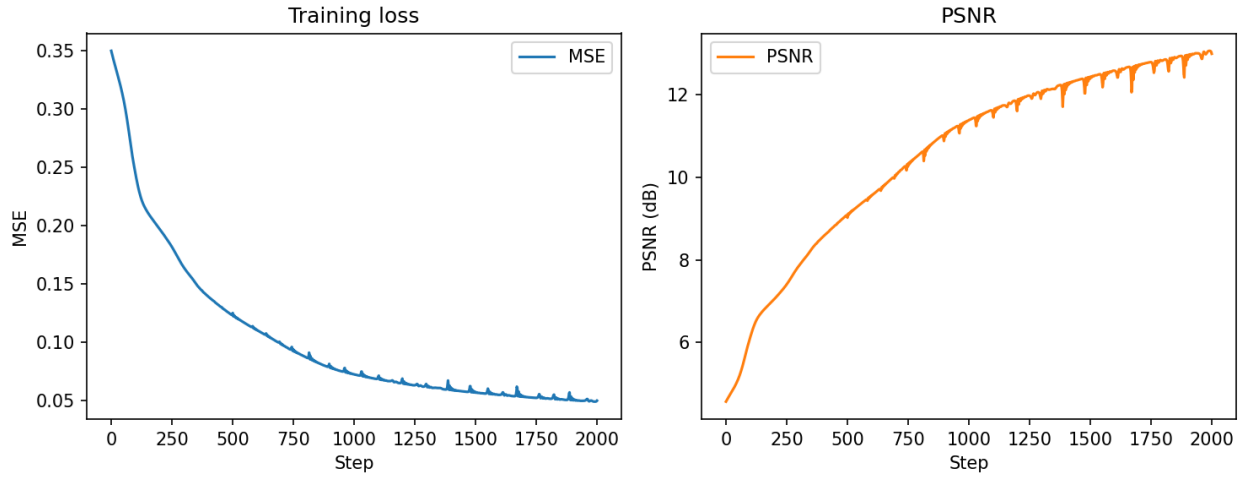


Figure 4: M3 metrics at step 1999

M4 — Tanh activation (8.62 dB, -1.84 vs M1) Although tanh is smooth (which could lead us to think it would provide better gradients compared to ReLU activations), we see that it perform significantly worse. This is probably due to the fact it saturates at high values, leading to very small gradients.

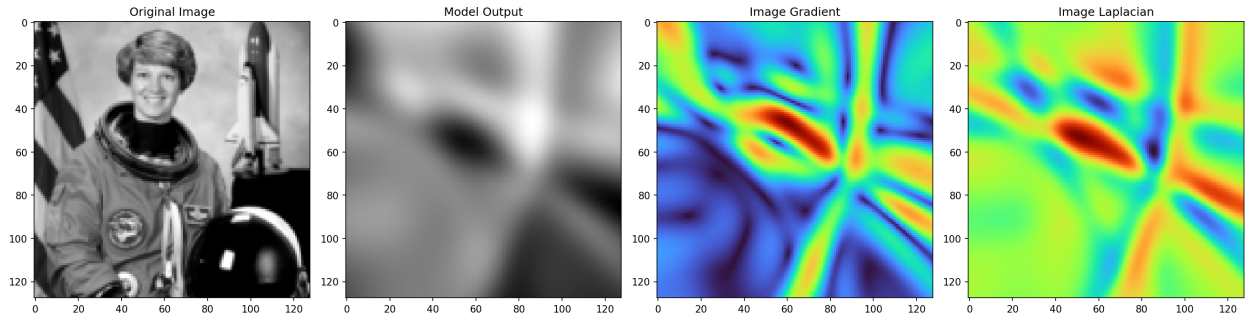


Figure 5: M4 field at step 1999

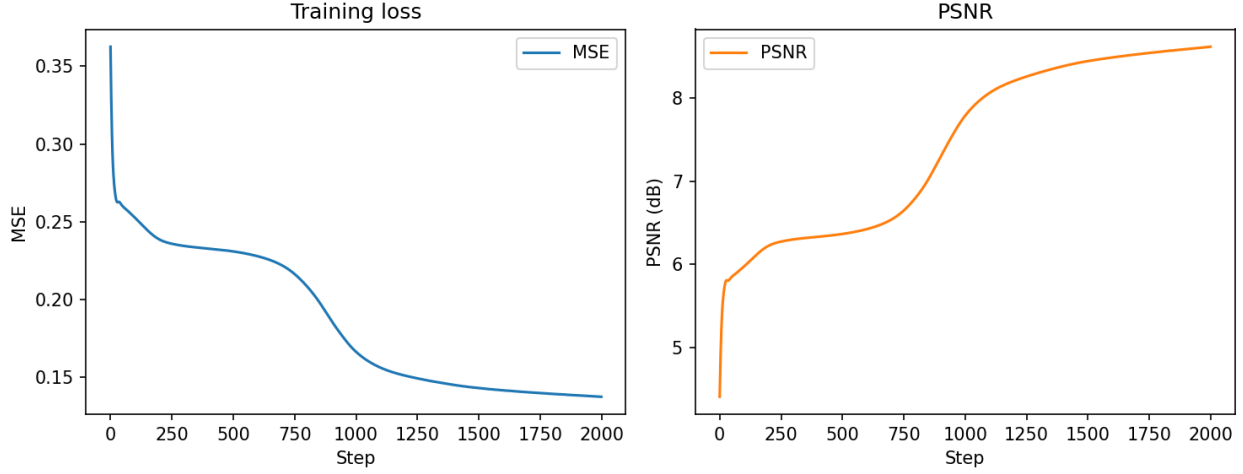


Figure 6: M4 metrics at step 1999

M5 — Higher LR (12.31 dB, +1.85 vs M1) With only 2000 steps, 10^{-3} takes larger Adam steps and reduces MSE faster than 10^{-4} before training stops. This is mainly **more optimization progress in a fixed budget**, not a better architecture. Curves show PSNR still climbing at the end.

M6 — Lower LR (6.72 dB, -3.74 vs M1) 10^{-5} moves weights too slowly: after 2000 steps the network is still **underfit**, especially on high frequencies. Loss and PSNR curves are flatter than M1, consistent with not having converged in the allotted steps.

2. SIREN sweep

Bold entries differ from the S1 baseline (128 features, $\omega_0 = 20/20$, linear last layer, $\text{lr} = 10^{-4}$).

Run	Width	ω_0 (1st layer)	ω_0 (hidden layers)	Linear last layer	LR	Notes	PSNR (dB)
S1	128	20	20	True	10^{-4}	Baseline (linear last layer)	35.01
S2	128	20	20	False	10^{-4}	Sine last layer (vs S1)	27.41
S3	128	5	5	True	10^{-4}	Low ω_0	19.09
S4	128	30	30	True	10^{-4}	High ω_0	41.81
S5	128	30	30	False	10^{-4}	High ω_0 + sine last (vs S4)	31.32
S6	128	20	10	True	10^{-4}	Split ω	30.62
S7	256	20	20	True	10^{-4}	Wider	40.27
S8	128	20	20	True	10^{-5}	LR low	21.76
S9	128	20	20	True	5×10^{-4}	LR high	43.52
S10	128	10	10	True	10^{-4}	Mid ω_0	24.72

S1 — Baseline (35.01 dB) Paper-style SIREN (linear last layer, $\omega_0 = 20$) already far exceeds the best MLP (M3 at 13.00 dB): reconstruction is sharper and the gradient/Laplacian panels align more closely with image structure.

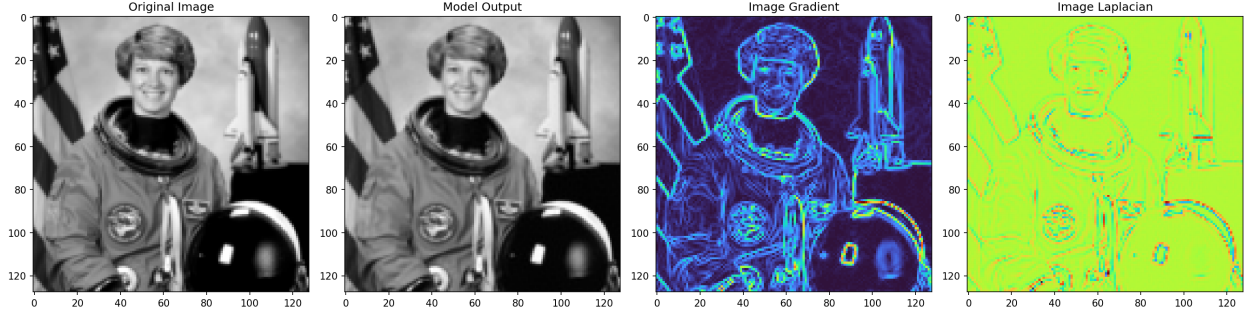


Figure 7: S1 field at step 1999

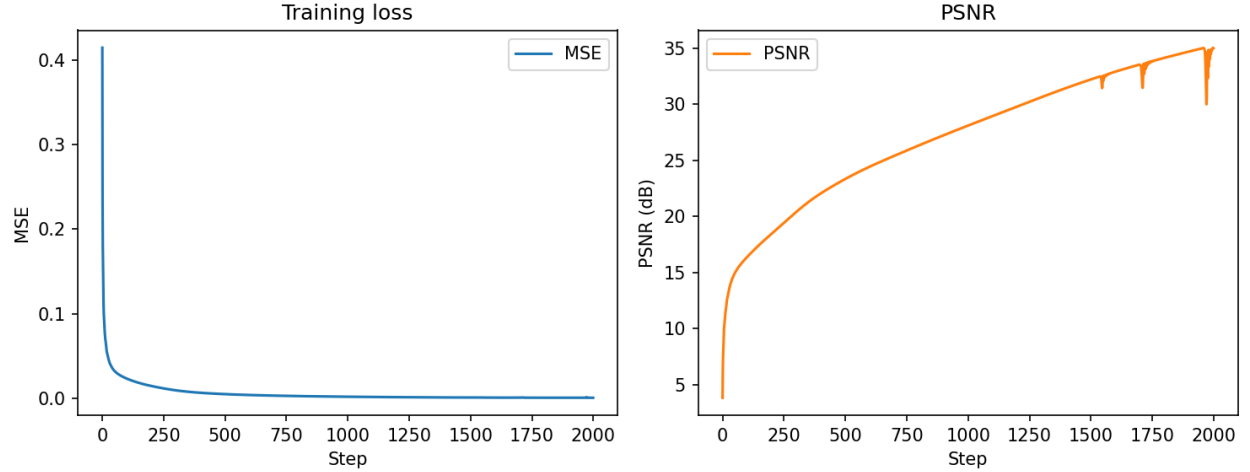


Figure 8: S1 metrics at step 1999

S2 — Sine last layer (27.41 dB, -7.60 vs S1) Matching ω_0 but using a sine output layer constrains pixel values and changes how high frequencies are composed at the head, which hurts PSNR relative to the linear last layer in S1. We can also observe some weird artifacts in the image gradient generated, and the Laplacian contains almost no signal.

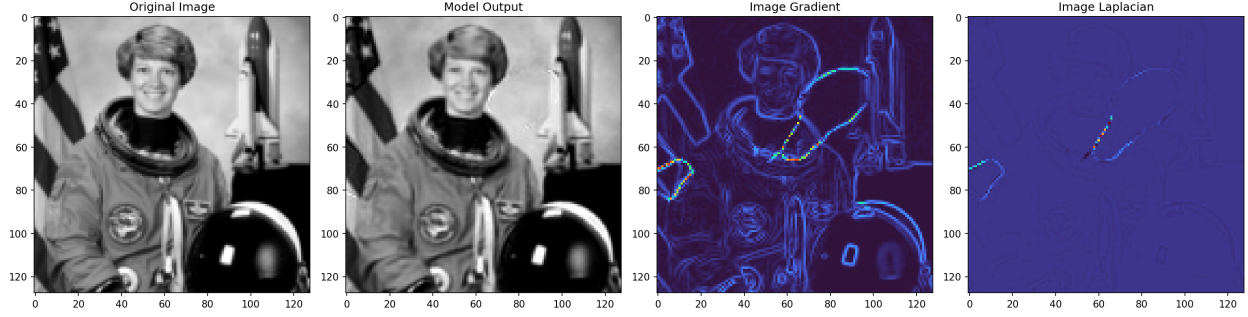


Figure 9: S2 field at step 1999

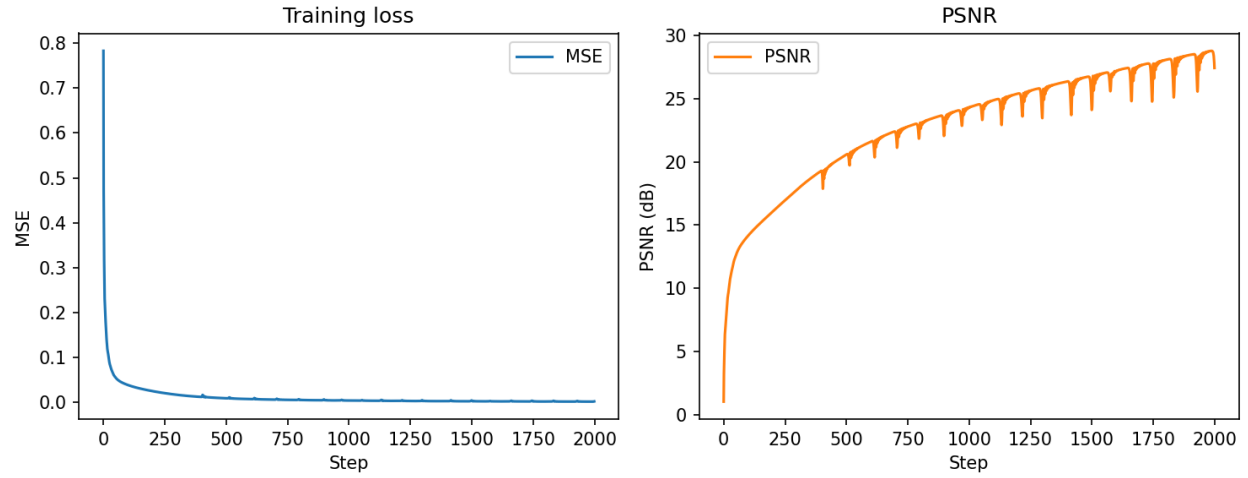


Figure 10: S2 metrics at step 1999

S3 — Low ω_0 (19.09 dB, -15.93 vs S1) With $\omega_0 = 5$, the network favors very low spatial frequencies and produces an overly smooth fit—similar in spirit to spectral bias in MLPs, and the weakest SIREN in the sweep. However, it still provides a fitting for the image Laplacian, which MLP models do not.

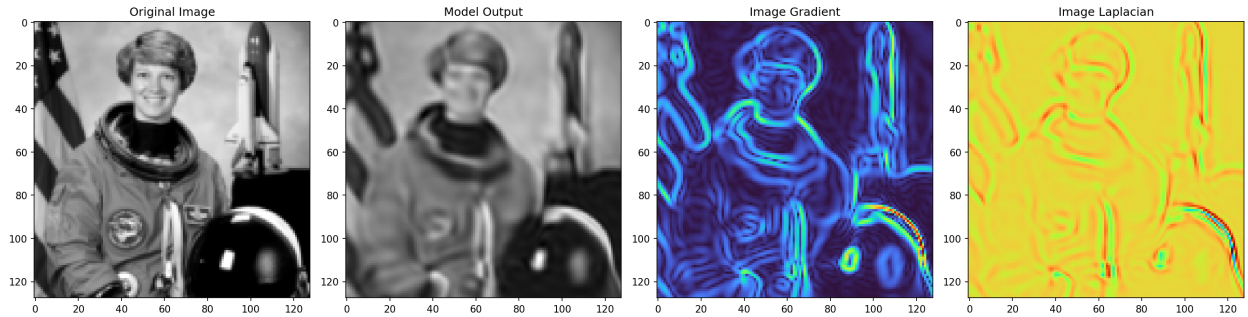


Figure 11: S3 field at step 1999

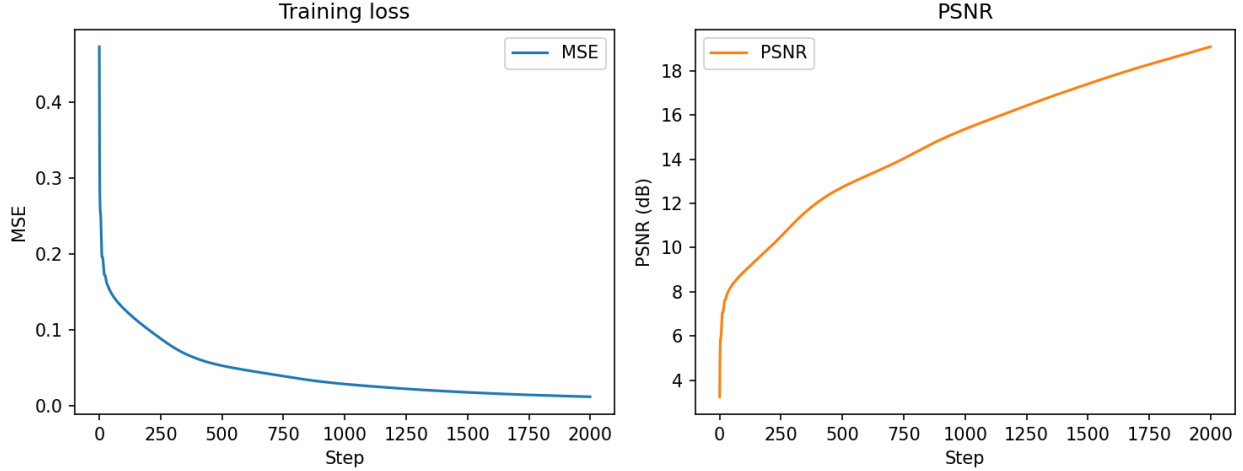


Figure 12: S3 metrics at step 1999

S4 — High ω_0 (41.81 dB, +6.80 vs S1) Raising ω_0 to 30 increases the band of representable frequencies, improving edges and fine detail. PSNR gains are substantial.

S5 — High ω_0 + sine last (31.32 dB, -10.49 vs S4) Same high ω_0 as S4 but with a sine last layer: performance drops sharply, strengthening the paper’s choice for linear last layer.

S6 — Split ω (30.62 dB, -4.40 vs S1) Using $\omega_0 = 20$ in the first layer and 10 in hidden layers misaligns frequency bandwidth across depth and underperforms a uniform $\omega_0 = 20$. Intuitively, the first layer lifts coordinates into a relatively high-frequency feature space, but later layers with smaller ω_0 can only mix those features with slower sinusoids—so fine detail introduced early is damped before it reaches the output.

S7 — Wider (40.27 dB, +5.26 vs S1) 256 hidden features add capacity for composing sinusoids over (x, y) and give a large PSNR boost, nearly matching S4 without changing ω_0 or LR.

S8 — Lower LR (21.76 dB, -13.25 vs S1) At $\text{lr} = 10^{-5}$ the model underfits within 2000 steps, analogous to M6: optimization budget, not architecture, is the bottleneck.

S9 — Higher LR (43.52 dB, +8.50 vs S1) Best SIREN in the sweep. $\text{lr} = 5 \times 10^{-4}$ converges faster and reaches the highest PSNR; metrics suggest the run is still improving at step 1999, so more steps or a schedule could help further.

Training progression (S9). Field visualizations below show how reconstruction, image gradient, and Laplacian evolve over training (summaries every 500 steps). PSNR rises from 3.61 dB at step 0 to 29.0 dB at 500, 34.8 dB at 1000, 40.1 dB at 1500, and 43.5 dB at 1999—coarse structure appears early, with fine detail and sharper derivative panels emerging in later steps.

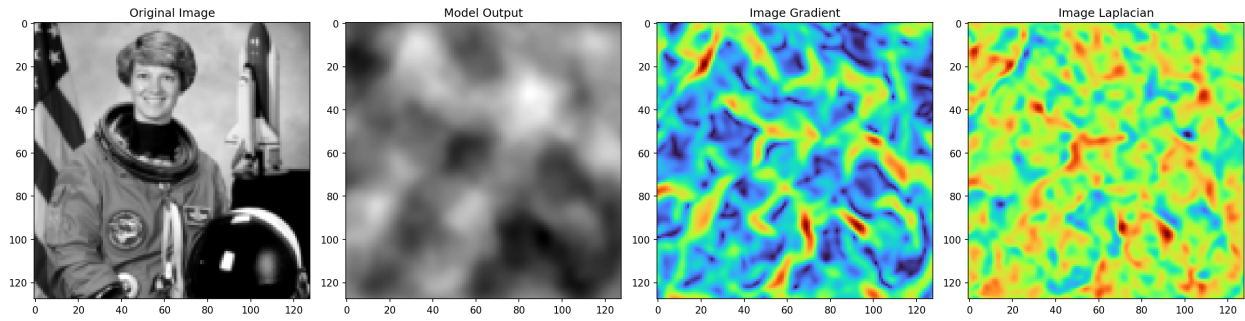


Figure 13: S9 field at step 0

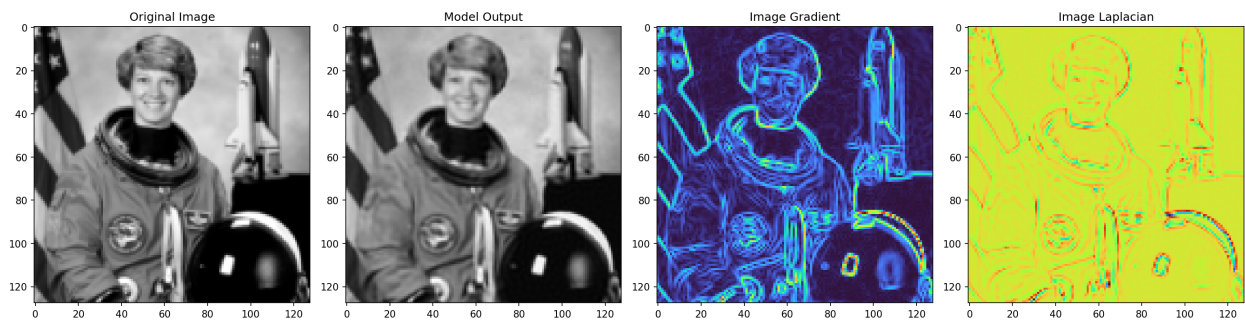


Figure 14: S9 field at step 500

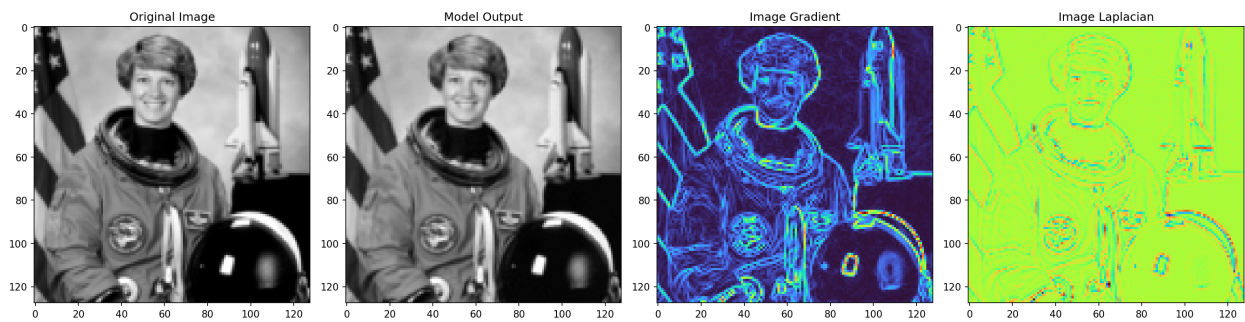


Figure 15: S9 field at step 1000

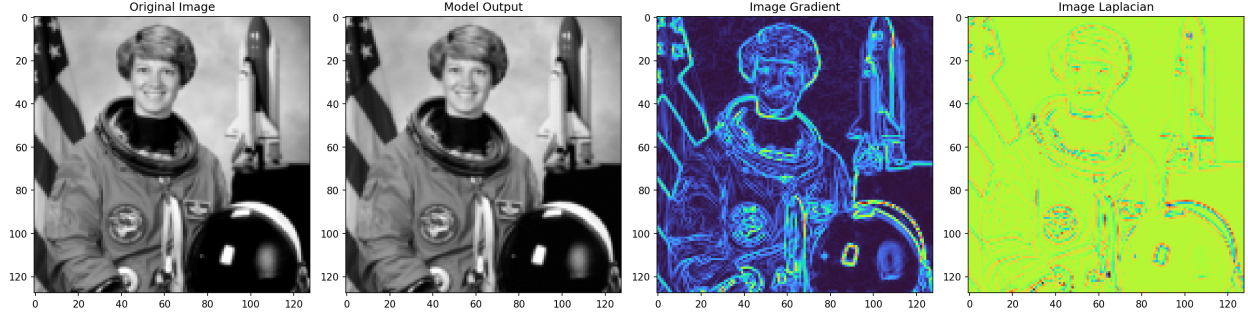


Figure 16: S9 field at step 1500

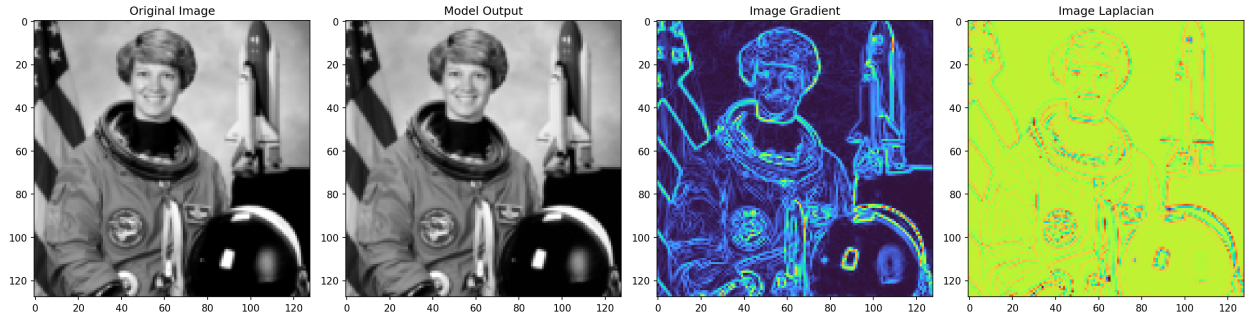


Figure 17: S9 field at step 1999

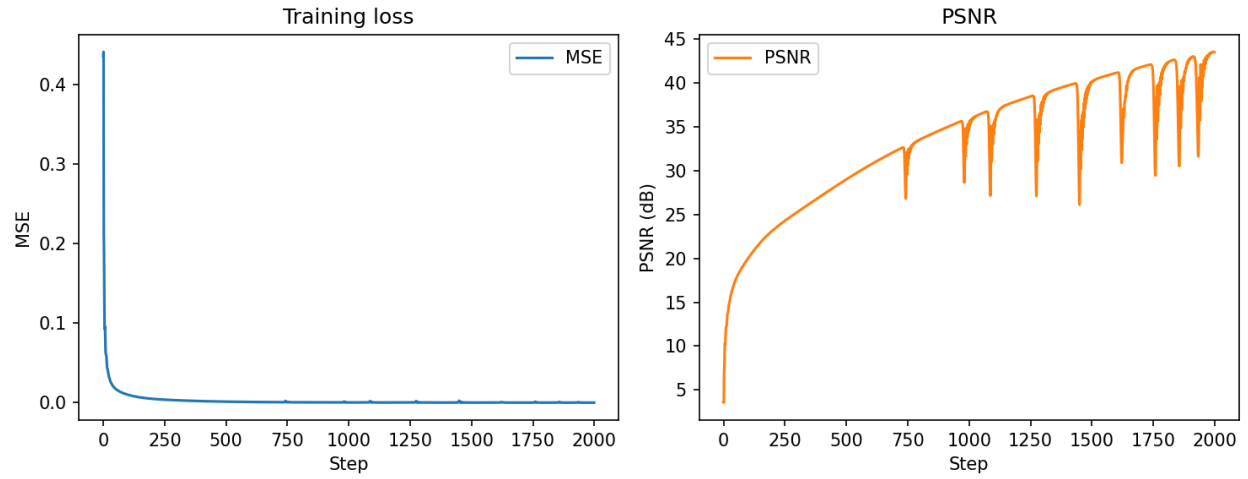


Figure 18: S9 metrics at step 1999

S10 — Mid ω_0 (24.72 dB, -10.29 vs S1) Across S3, S10, S1, and S4, PSNR rises as ω_0 increases (19.1 $\rightarrow$ 24.7 $\rightarrow$ 35.0 $\rightarrow$ 41.8 dB), so **higher** ω_0 helps this image fit finer spatial detail. S10 is better than the very low setting in S3 but still far below $\omega_0 = 20$ or 30—there is no safe “default in the middle”; ω_0 should be chosen toward the high end of what still trains stably.

Sweep comparison (final PSNR)

Side-by-side bar charts for all MLP (M1–M6) and SIREN (S1–S10) configurations at final step (1999). SIREN reaches much higher PSNR than any MLP run; within each family, capacity and ω_0 /LR tuning dominate the spread.

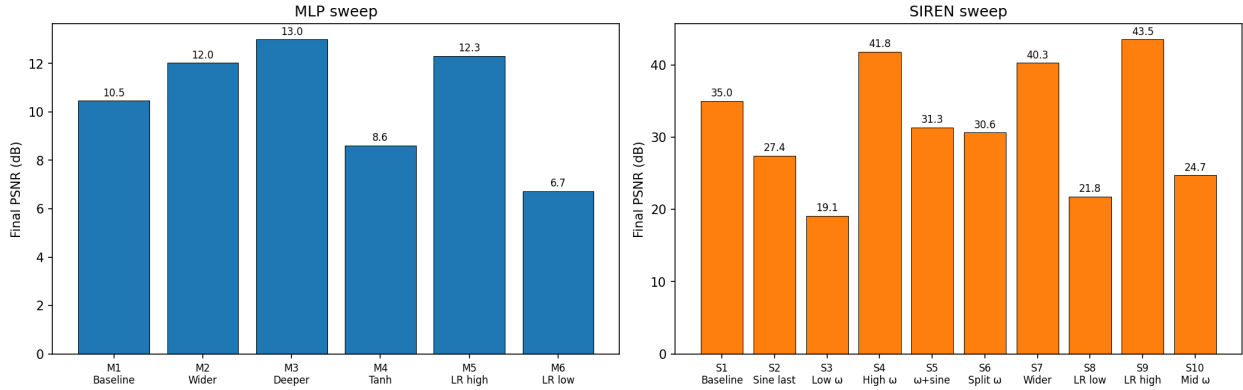


Figure 19: MLP and SIREN sweep final PSNR

C. Optimal SIREN configuration

After benchmarking 16 MLP/SIREN variants on the `astronaut` image (128×128), our best configuration is **run S9**:

Setting	Value
Model	SIREN
Image	<code>astronaut</code> (128×128)
Hidden width	128
Hidden layers	3
ω_0 (first layer)	20
ω_0 (hidden layers)	20
Last layer	Linear (<code>last_layer_linear=True</code>)
Learning rate	5×10^{-4}
Optimizer	Adam
Loss	MSE
Training steps	2000
Final PSNR	43.52 dB

This run outperformed the S1 baseline (+8.50 dB) and all other sweep configurations, including wider networks (S7) and higher ω_0 (S4). The main gain came from a higher learning rate that made better use of the fixed 2000-step budget; see the S9 training progression above for how reconstruction quality improves over time.